

An attempt to connect RTC and SD card to FM-7 (RTC section)

 flexonsbd.blogspot.com/2023/03/fm-7rtcsdrtc.html

Thursday, March 9, 2023

An attempt to connect an RTC and SD card to the FM-7 (RTC section)

I made an RTC & SD card board for the FM-7.

[Correction] The OneDrive link was incorrect, so it has been changed to the correct link.

Previously, I introduced an attempt to use the 6821PIA to read and write to an SD card via I2C and I2C.

"Successfully established I2C communication between PIA and RTC using assembly language (October 27, 2018)"

(<https://flexonsbd.blogspot.com/2018/10/piartci2c.html>)

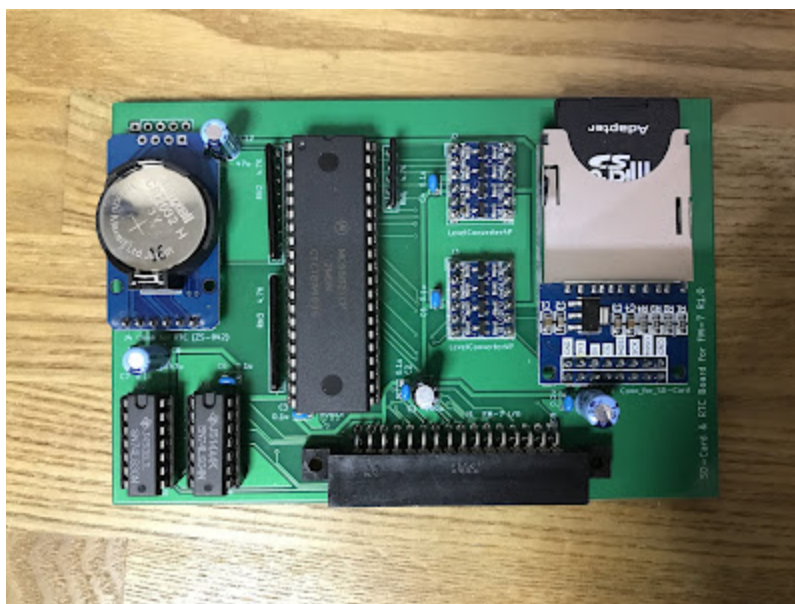
"Trying to read and write to an SD card using 6809 assembler (June 12, 2022)"

(<https://flexonsbd.blogspot.com/2022/06/sd6809.html>)

The ultimate goal is to implement an RTC (Real-Time Clock) and an SD card into the FM-7, but while the RTC is relatively easy, implementing an SD card is not so simple.

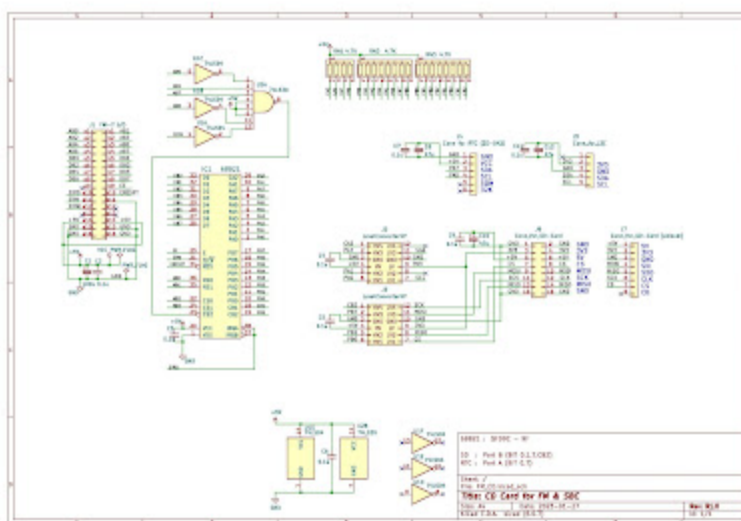
However, I realized that just thinking about it wouldn't get me anywhere, so I decided to try making a circuit board first.

This is the circuit board I created.



Fabricated RTC & SD card board

Here is the circuit diagram.



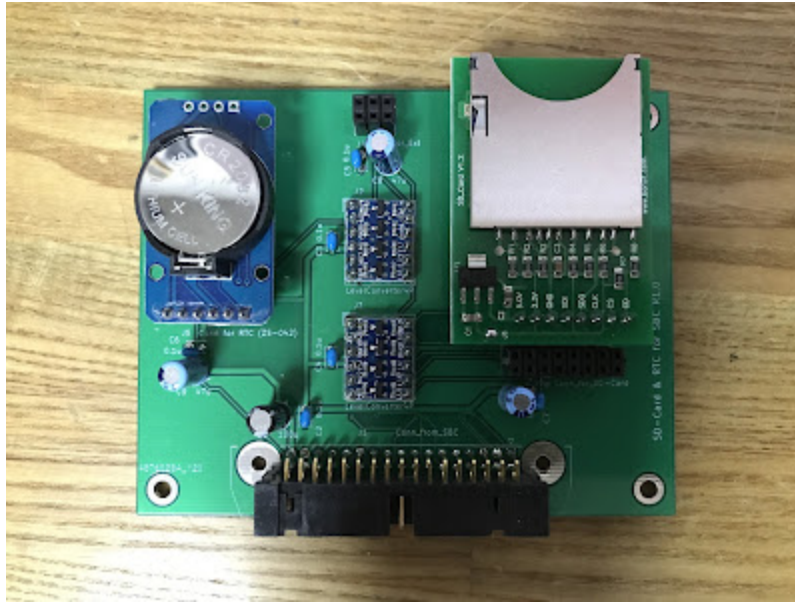
Circuit diagram

Basically, it's the same as the one I introduced before, using the 68B21's A port to control the RTC via I2C and the B port to control the SD card via SPI . Note that the RTC is 5V powered, so it's directly connected to the 68B21, but the SD card is 3.3V powered, so it goes through a level conversion module, and the 3.3V power for the SD card is supplied from the SD card.

The RTC board used is the ZS-042, which uses a DS3231 chip, and the traces have been cut to prevent battery charging. In addition to the SD card board shown in the image, traces for both types of boards are provided so that a board previously sold by Akizuki Denshi can also be used. (It is mounted on the microcontroller board below.)

At the same time, I also manufactured circuit boards for my custom-made 6809/6802 microcontroller boards.

This circuit board is used to automatically read the date when Flex9 starts up.



Microcontroller board

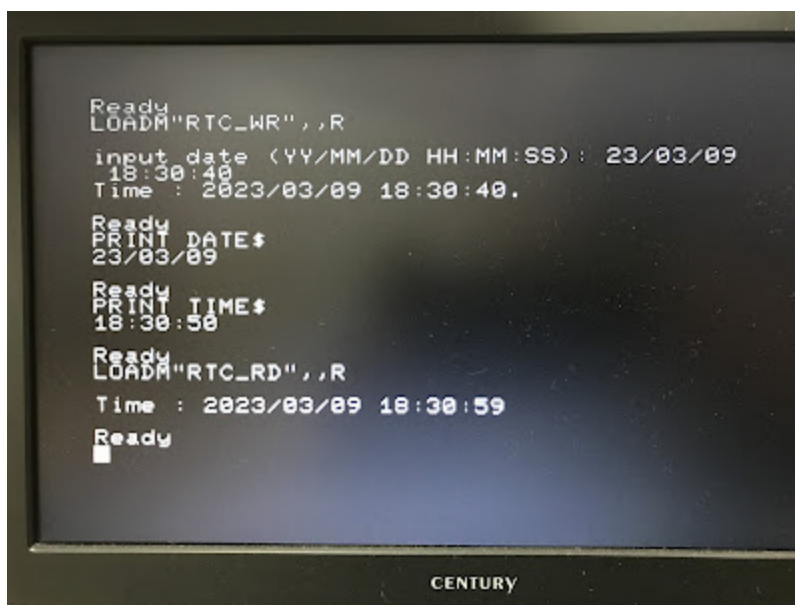
This time, I will only introduce the RTC (Real-Time Clock) part.

This is how it looks when attached to an FM-7.



Scene of operation

This is how it looks while it's running.



Execution screen

The software I created for F-BASIC V3.0 on the FM-7 consists of two programs: RTC_RD, which reads from the RTC, and RTC_WR, which writes to it. RTC_WR directly writes the date and time to the F-BASIC work area, so RTC_WR is exclusive to V3.0.

As you can see in the image, the data that was written is displayed using the PRINT statement, confirming that it has been set correctly.

For reference, here is a list of RTC_WRs.

```

0001 *          0002 *   RTC_WR . CMD for F - BASIC V3 . 0          0003 *
0004 *   WRITE DATE to RTC3231 by I2C on F - BASIC          0005 *
0006 * 6821 ( PIA ) Port A <--> RTC3231          0007 *          0008
* Applied to F - BASIC   '2023.03.09          0009 * Original version '
2018.11 . 030010 * Copyright by Y. Yamamoto          0011 *
0012   FBB8 INCHA EQU $FBB8          0013   FBBB OUTCHA EQU $FBBB          0014
E2E9 SETIME EQU $E2E9          set time data routine0015 02FA          WKDATE EQU
$02FA work area for date          0016          0017   FD9C PADAT EQU $FD9C
0018   FD9D PACTL EQU PADAT + 1          0019          0020 000D          CR
EQU $0D          0021 000A          LF EQU $0A          0022
0023 00D0          SADRW EQU $D0          ( 01101000 + 0 )          0024 00D1
SADRR EQU $D1          ( 01101000 + 1 )          0025          0026 6000
ORG $6000          0027          0028 6000 20 03          START BRA MAIN
0029 6002 01          VN FCB          1          0030 6003          FF ODAT FCB $FF
0031 6004 00          WK FCB          00          0032          0033 6005
MAIN EQU          *          0034 6005          BD          619A          JSR PCRLF
0035 * get date data          0036 6008 8E 61CC          LDX          #MDAT
0037 600B          BD          618A          JSR PSTRNG          0038
0039 600E 8E 621F          LDX          #INBUF+20          0040 6011 4F
CLRA          0041 6012          C6          14          LDB          #20
0042 6014          A7          82          CLLOP STA          , -X          0043 6016 5A
DECB          0044 6017 26          FB BNE CLLOP          0045 6019
BD FBB8 INDLOP JSR INCHA          0046 601C          BD FBBB JSR OUTCHA
0047 601F          A7          80          STA          , X +          0048 6021 81 0D
CMPA          #CR          0049 6023 26          F4 BNE INDLOP
0050 6025 6F 1F          CLR          - 1 , X          0051 6027 8E
620B          LDX          #INBUF          0052          0053 602A
C6          07          LDB          #7          0054 602C 108E 61FE
LDY          #WRDATA          0055 6030 34 04          INLOP PSHS B
0056 6032          EC          81          LDD          , X ++          0057 * if B is
Sep . not 2dec          0058 6034          C1          30          CMPB          #'0
0059 6036 25 04          BLO INL1          0060 6038          C1          39
CMPB          #'9          0061 603A 23 06          BLS INL2
0062 603C 1F 89          INL1 TFR A , B          0063 603E 86 30
LDA          #'0          0064 6040 30 1F          LEAX          - 1 , X
0065 6042 48          INL2 ASLA          0066 6043 48
ASLA          0067 6044 48          ASLA          0069 6046          C4          0F
0068 6045 48          ASLA          0070 6048          F7          6004          STB WK
ANDB          #$0F          0071 604B          BB          6004          ADDA WK          0072 604E          A7 A0
STA          , Y +          0073 6050 30 01          LEAX          1 , X skip
separator          0074 6052 35 04          PULS B
0075 6054 5A          DECB          0076 6055 27 0A
BEQ INEND          0077 6057          C1          04          CMPB          #4

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0078 6059 26      D5 BNE INLOP                                0079 605B 86 01      LDA
#$01 add week data                                0080 605D      A7 A0 STA      , Y +      0081
605F 20      CF BRA INLOP                                0082 6061      INEND EQU      *
0083                                0084 * reset pia                                0085 * port A , B : output , data
$ff : out                                0086 6061 4F      CLRA
0087 6062      B7 FD9D STA PACTL      select ddra      0088 6065 86      FF LDA
#$FF                                0089 6067      B7 FD9C STA PADAT      set output mode
0090 606A 86 36      LDA      #$36                                0091 606C      B7
FD9D STA PACTL      select peripheral      0092 * out High = bus release ( set
input mode )      0093 606F 86      FF LDA      #$FF
0094 6071      B7      6003      STA 0DAT save out data      0095 6074      B7
FD9C STA PADAT      0096                                0097 * start cond . ( scl = H , sda
= H -> L )      0098 6077 17 00B7      LBSR SETDTL
0099 607A 17 0096      LBSR SETCKL      0100
0101 * send slave adr + W ( $D0 )      0102 607D 86      D0 LDA      #SADRW
0103 607F 17 00CD      LBSR WRDAT      0104
0105 * wait ACK      0106 6082 17 00E5      LBSR WACK
0107                                0108 * send reg top adr ( $00 )      0109 6085 86 00
LDA      #$00 reg adr:00-06      0110 6087 17 00C5      LBSR WRDAT
0111                                0112      *wait ACK0113      608A      17      00DD
LBSR      WACK      0114                                0115      *write data0116
608D      8E      6205      LDX      #WRDATA+7      0117      6090      C6      07
LDB      #7      0118      6092      34      04      WRLOP      PSHS      B
0119      6094      A6      82      LDA      , -X      0120      6096      17
00B6      LBSR      WRDAT      0121      6099      17      00CE      LBSR
WACK      0122      609C      35      04      PULS      B      0123
609E      5A      DECB      0124      609F      26      F1
BNE      WRLOP      0125      0126      *stop0127
*scl=H,sda=L->H0128      60A1      8D      7F      BSR      SETCKH      0129
60A3      17      009A      LBSR      SETDTH      0130      0131
*display data0132      60A6      BD      619A      JSR      PCRLF      0133
60A9      8E      61ED      LDX      #MDATE      0134      60AC      17      00D1
LBSR      DSPMSG      0135      60AF      108E      61F7      LDY      #MDELM
0136      60B3      8E      61FE      LDX      #WRDATA      year month day week
hour min sec0137      60B6      8C      6201      DLOP      CMPX      #WRDATA+3      0138
60B9      27      08      BEQ      DLOP2      0139      60BB      BD      61A9
JSR      OUTHEX      0140      60BE      A6      A4      LDA      ,Y
0141      60C0      BD      FBFB      JSR      OUTCHA      0142      60C3      31
21      DLOP2      LEAY      1,Y      0143      60C5      30      01      LEAX
1,X      0144      60C7      8C      6205      CMPX      #WRDATA+7      0145
60CA      26      EA      BNE      DLOP      0146      60CC      BD      619A
JSR      PCRLF      0147      0148      *set time to F-
BASIC0149      *set to work area using F-BASIC routine0150      60CF      CE
6205      LDU      #WKTIME      0151      60D2      8E      6202      LDX
#WRDATA+4      0152      60D5      C6      03      LDB      #3      0153
60D7      34      04      STLOP      PSHS      B      0154      60D9      A6      80
LDA      ,X+      0155      60DB      1F      89      TFR      A,B
0156      60DD      44      LSRA      0157      60DE      44
LSRA      0158      60DF      44      LSRA      0159

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60E0    44                LSRA                0160    60E1    A7    C0
STA      ,U+                0161    60E3    1F    98                TFR      B,A
0162    60E5    84    0F                ANDA    #$0F                0163    60E7    A7    C0
STA      ,U+                0164    60E9    35    04                PULS      B
0165    60EB    5A                DECB                0166    60EC    26    E9
BNE      STL0P                0167    60EE    33    5A                LEAU      -6,U
0168    60F0    BD    E2E9                JSR      SETIME                0169
*set date to F-BASIC0170                *set to work area directly0171    60F3    CE
02FA                LDU      #WKDATE                0172    60F6    8E    61FE                LDX
#WRDATA                0173    60F9    C6    03                LDB      #3                0174
60FB    34    04    SDLOP    PSHS    B                0175    60FD    A6    80
LDA      ,X+                0176    60FF    1F    89                TFR      A,B
0177    6101    44                LSRA                0178    6102    44
LSRA                0179    6103    44                LSRA                0180
6104    44                LSRA                0181    6105    A7    C0
STA      ,U+                0182    6107    1F    98                TFR      B,A
0183    6109    84    0F                ANDA    #$0F                0184    610B    A7    C0
STA      ,U+                0185    610D    35    04                PULS      B
0186    610F    5A                DECB                0187    6110    26    E9
BNE      SDLOP                0188    6112    39                RTS                0189
0190                *****0191                *set
clk low(out b0=0)0192    6113                SETCKL    EQU    *                0193
6113    34    04                PSHS    B                0194    6115    F6    6003
LDB      ODAT                0195    6118    C4    FE                ANDB    #$FE
b0=00196    611A    F7    FD9C                STB      PADAT                0197    611D    F7
6003                STB      ODAT                0198    6120    35    84                PULS
B,PC                0199                0200 * set clk Hi = bus release ( set input mode )
0201 6122                SETCKH EQU    *                0202 6122 34 04
PSHS B                0203 6124    F6    6003                LDB ODAT
0204 6127    CA    01                ORB      #$01 b0=1    0205 6129    F7 FD9C STB
PADAT                0206 612C    F7    6003                STB ODAT
0207 612F 35 84                PULS B , PC                0208
0209 * set data low ( out b7 = 0 )                0210 6131                SETDTL EQU
*                0211 6131 34 04                PSHS B
0212 6133    F6    6003                LDB ODAT                0213 6136    C4
7F                ANDB    #$7F b7=0    0214 6138    F7 FD9C STB PADAT
0215 613B    F7    6003                STB ODAT                0216 613E 35 84
PULS B , PC                0217                0218 * set data Hi = bus
release ( set input mode )                0219 6140                SETDTH EQU    *
0220 6140 34 04                PSHS B                0221 6142    F6
6003                LDB ODAT                0222 6145    CA    80                ORB
#$80 b7=1    0223 6147    F7 FD9C STB PADAT                0224 614A    F7    6003
STB ODAT                0225 614D 35 84                PULS B , PC
0226                0227 * write data or ADR                0228 * A : data or adr
0229 614F                WRDAT EQU    *                0230 614F    C6    08
LDB      #8                0231 6151 8D    C0 WLOP BSR SETCKL                0232
6153 34 04                PSHS B                0233 6155 48
ASLA b7 -> C                0234 6156 24 04                BCC WL1                0235
6158 8D    E6 BSR SETDTH                0236 615A 20 02                BRA WL2

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0237 615C 8D      D3 WL1 BSR SETDTL      0238 615E 8D      C2 WL2 BSR SETCKH
0239 6160 8D      B1 BSR SETCKL          0240 6162 8D      CD BSR SETDTL
0241 6164 35 04          PULS B          0242 6166 5A
DECB          0243 6167 26      E8 BNE WLOP          0244 6169
39          RTS          0245          0246 * wait ACK
0247 * ACK : sda = L          0248 616A          WACK EQU      *
0249 616A 8D      B6 BSR SETCKH          0250 616C      F6 FD9C WALOP LDB PADAT
0251 616F 58          ASLB data -> C          0252 6170 25      FA BCS WALOP
0253 6172 8D 9F          BSR SETCKL          0254 6174 8D      CA BSR
SETDTH sda release          0255 6176 39          RTS
0256          0257 * set ACK          0258 6177          SACK EQU
*          0259 6177 8D      B8 BSR SETDTL          0260 6179 8D
A7 BSR SETCKH          0261 617B 8D 96          BSR SETCKL
0262 617D 8D      C1 BSR SETDTH sda release          0263 617F 39          RTS
0264          0265 * disp message          0266 6180          DSPMSG
EQU      *          0267 6180      A6      80      DSPL0P LDA      , X +
0268 6182 27 05          BEQ DSPEND          0269 6184      BD FBBB
JSR OUTCHA          0270 6187 20      F7 BRA DSPL0P          0271 6189 39
DSPEND RTS          0272          0273 * pstring
0274 618A          PSTRNG EQU      *          0275 618A 34 02
PSHS A          0276 618C      A6      80      PSLOP LDA      , X +
0277 618E 81 04          CMPA      #4          0278 6190 27 05
BEQ PSEND          0279 6192      BD FBBB JSR OUTCHA          0280
6195 20      F5 BRA PSLOP          0281 6197 35 02      PSEND PULS A
0282 6199 39          RTS          0283          0284 *
print CR , LF          0285 619A          PCRLF EQU      *
0286 619A 34 02          PSHS A          0287 619C 86 0D
LDA      #CR          0288 619E      BD FBBB JSR OUTCHA
0289 61A1 86 0A          LDA      #LF          0290 61A3      BD
FBBB JSR OUTCHA          0291 61A6 35 02          PULS A
0292 61A8 39          RTS          0293          0294 *
print hex          0295 61A9          OUTHEX EQU      *
0296 61A9 34 06          PSHS A , B          0297 61AB      A6
84          LDA      , X          0298 61AD 1F 89          TFR A , B
0299 61AF 44          LSRA          0300 61B0 44
LSRA          0301 61B1 44          LSRA
0302 61B2 44          LSRA          0303 61B3 8D 0E
BSR OUTS          0304 61B5      BD FBBB JSR OUTCHA          0305
61B8 1F 98          TFR B , A          0306 61BA 84 0F
ANDA      #$0F          0307 61BC 8D 05          BSR OUTS
0308 61BE      BD FBBB JSR OUTCHA          0309 61C1 35 86          PULS A , B
, PC          0310          0311 61C3 8B 30          OUTS ADDA      #$30
0312 61C5 81 39          CMPA      #$39          0313 61C7 23 02
BLS OUTSE          0314 61C9 8B 07          ADDA      #$7
0315 61CB 39          OUTSE RTS          0316          0317 *
0318 61CC 69          MDAT FCC      "input date (YY/MM/DD HH:MM:SS): "
0319 61EC 04          FCB      4          0320 61ED 54
MDATE FCC      / Time : 20 /          0321 61F6 00          FCB
0          0322 61F7 2F          MDELM FCC      "// : : ."

```

```

0323 *****
61FE      FF WRDATA RMB      7
0327              0328 620B
0330 END          START
detected .      Label      addressCLLOP      6014      CR      000D      DLOP      60B6
DLOP2      60C3      DSPEND      6189DSPL0P      6180      DSPMSG      6180      INBUF
620B      INCHA FBB8 INDLOP      6019INEND      6061      INL1      603C      INL2
6042      INLOP      6030      LF      000AMAIN      6005      MDAT      61CC
MDATE      61ED      MDELM      61F7      ODAT      6003OUTCHA FBBB OUTHEX      61A9
OUTS      61C3      OUTSE      61CB      PACTL FD9DPADAT FD9C PCRLF      619A      PSEND
6197      PSLOP      618C      PSTRNG      618ASACK      6177      SADRR      00D1
SADRW      00D0      SDLOP      60FB      SETCKH      6122SETCKL      6113      SETDTH
6140      SETDTL      6131      SETIME E2E9 START      6000STLOP      60D7      VN
6002      WACK      616A      WALOP      616C      WK      6004WKDATE      02FA
WKTIME      6205      WL1      615C      WL2      615E      WLOP      6151WRDAT
614F      WRDATA      61FE      WRL0P      6092

```

By executing RTC_WR when F-BASIC starts to set the date and time, you can then access the date and time from F-BASIC.

With the above steps, we have successfully installed an RTC in the FM-7, but the problem is the SD card.

If you want to save and read files to and from an SD card using a proprietary format, I think this circuit board I created might work. However, if you want to use a format common to Windows PCs (FAT16/32), it seems impossible with just the FM-7 and 6821 machine code. Even after searching online, it seems that most of the FAT libraries available are those designed for microcontrollers.

However, although it may be difficult, I would like to try to do as much as possible without using a microcontroller.

If I manage to successfully read and write files, I will share the solution.

(Even after searching online, I can't seem to find any articles about using SD cards in FM-series radios. Or am I just missing them? All I seem to find are articles about 80-series radios...)

I've uploaded the source code and binaries for the two software programs (RTC_WR and RTC_RD) I created for the RTC to [OneDrive](#) . Note that the assembly code for these source files uses my previously introduced cross-assembler, 6809AsmWin.exe.